

Christopher Felder

Curriculum Vitae

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Employment

- 2025 – **Assistant Professor.**
University of South Florida, Department of Mathematics & Statistics.
- 2022 – 2025 **Zorn Postdoctoral Fellow.**
Indiana University, Bloomington, Department of Mathematics.

Education

- 2022 **PhD., Mathematics.**
Washington University in St. Louis.
Advisor: John E. M^cCarthy.
- 2018 **A.M., Mathematics.**
Washington University in St. Louis.
- 2016 **B.S., Mathematics.**
Butler University. Indianapolis, Indiana.
- 2016 **B.S., Mechanical Engineering.**
Indiana University–Purdue University. Indianapolis, Indiana.

Grants

- 2026-2028 **AMS-Simons Travel Grant.**
(\$6,000).
- 2026 **Faculty International Travel Grant, USF.**
Funding for mini course at the Institute of Mathematics of the Jagiellonian University (\$2,500).
- 2018 **ICM Travel Grant.**
AMS administered NSF grant for travel to the ICM in Rio de Janeiro, Brazil (\$3,300).
- 2015 **Seitz Award.**
Butler University endowed grant for international research travel (\$5,300);
Used for 8 week stay at Universidad Complutense de Madrid.

Awards

- 2025 **Max Zorn Teaching Award.**
Awarded by the Department of Mathematics at Indiana University in recognition of excellence in the teaching of mathematics (\$1,000).
- 2019 **Robert McDowell Teaching Award.**
Awarded by the Department of Mathematics at Washington University for excellence in the teaching of mathematics & outstanding performance as a teaching assistant.
- 2016 **Amos Carpenter Award.**
Awarded by the Department of Mathematics at Butler University for excellence in undergraduate mathematics research.
- 2015 **Bay Chotlos Award.**
Awarded by the Department of Mathematics at Butler University for dedication to a professional career in mathematics.

Preprints

- [1] **Forward operator monoids.**
Submitted (2024). arXiv:2409.16489 [↗](#).

Publications (Accepted/Published)

- [11] **Metric projections, zeros of optimal polynomial approximants, and some extremal problems in Hardy spaces.**
(w/ C. Bénéteau, R. Cheng, D. Khavinson, M. Manolaki, K. Maronikolakis)
Canad. J. Math. (2026). DOI [↗](#).
- [10] **Optimal polynomial approximants in H^p .**
(w/ R. Center and R. Cheng)
J. Math. Anal. Appl. (2025). DOI [↗](#).
- [9] **Some remarks on Shanks-type conjectures.**
Anal. Math. Phys. (2024). DOI [↗](#).

- [8] **More properties of optimal polynomial approximants in Hardy spaces.**
(w/ R. Cheng)
Pacific J. Math. (2024). DOI [↗](#).
- [7] **On interpolation by functions in ℓ_A^p .**
(w/ R. Cheng)
Contemp. Math. (Recent Progress in Function Theory and Operator Theory) (2024). DOI [↗](#).
- [6] **General optimal polynomial approximants, stabilization, and projections of unity.**
Anal. Theory Appl. (2023). DOI [↗](#).
- [5] **On the geometry of the multiplier space of ℓ_A^p .**
(w/ R. Cheng)
Concr. Oper. (2022). DOI [↗](#).
- [4] **Spectra for Toeplitz operators associated with a constrained subalgebra.**
(w/ D. Pfeffer and B. Russo)
Integral Equations Operator Theory (2022). DOI [↗](#).
- [3] **Convexity of the Berezin range.**
(w/ C. Cowen)
Linear Algebra Appl. (2022). DOI [↗](#).
- [2] **Analogues of finite Blaschke products as inner functions.**
(w/ T. Le)
Bull. Lond. Math. Soc. (2022). DOI [↗](#).
- [1] **Active Manifolds: A non-linear analogue to Active Subspaces.**
(w/ R. Bridges, A. Gruber, C. Hoff, K. Verma)
International Conference on Machine Learning (2019). DOI [↗](#).

Other Publications (Feature Stories)

- 2021 **Is Virtual Conferencing Here to Stay?**, *The Ampersand* [↗](#).
- 2020 **Meet Soumendra Lahiri, the new Stanley A. Sawyer Professor,** *The Ampersand* [↗](#).

- 2019 **A Calculated Move**, *The Ampersand* [↗](#).
- 2018 **The Politics of Polygons: Using Math to Determine Fairness**, *The Ampersand* [↗](#).

Teaching

At University of South Florida

- Spring 2026 **Linear Algebra**, MAS 3105.
Fall 2025 **Calculus II**, MAC 2312.

At Indiana University

- Spring 2025 **Elementary Computational Methods**, Math-M 371.
Spring 2025 **Calculus IV**, Math-M 312.
Fall 2024 **Calculus I**, Math-M 211.
Spring 2024 **Linear Algebra for Undergraduates**, Math-M 303.
Fall 2023 **Brief Survey of Calculus I**, Math-M 119.
Spring 2023 **Elementary Computational Methods**, Math-M 371.
Fall 2022 **Brief Survey of Calculus I**, Math-M 119.
Two sections.

At Washington University in St. Louis

- Summer 2021 **Elementary Probability and Statistics**, Math 2200, Co-instructor w/ Nathan Wagner [↗](#).
Summer 2021 **Introduction to Statistics**, Math 1011, Co-instructor w/ Nathan Wagner [↗](#).
Summer 2020 **Calculus II**, Math 132.
Two sections.
Summer 2019 **Matrix Algebra**, Math 309.

Invited Talks

- June 2026 **Barcelona Analysis Seminar**. Autonomia University of Barcelona, Spain.
June 2026 **Classical Analysis Seminar**. KU Leuven, Belgium.

- June 2026 **Workshop on Contemporary Topics in Complex Analysis.** Lund University, Sweden.
- May 2026 **University of La Laguna Mathematical Analysis and Applied Mathematics Seminar.** Tenerife, Spain.
- March 2026 **Complex (Analysis and Dynamics) II.** University of Texas at Dallas.
- February 2026 **Bench Math Session.** Mini course (two lectures). Jagiellonian University. Krakow, Poland.
- January 2026 **Joint Mathematics Meetings:** AMS Special Session on New Faces in Operator Theory. Washington, D.C.
- November 2025 **UTampa Math Lecture Series,** University of Tampa. Tampa, FL.
- October 2025 **AMS Fall Central Sectional Meeting.** Special Session on Real and Complex Function Theory with Application to Operator Theory. St. Louis University.
- October 2025 **Virginia Operator Theory and Complex Analysis Meeting.** University of Richmond.
- April 2025 **Vanderbilt University Analysis Seminar.** Nashville, TN.
- February 2025 **Colloquium,** University of South Florida. Tampa, FL
- February 2025 **Colloquium,** Colorado Mesa University. Grand Junction, CO
- January 2025 **Joint Mathematics Meetings:** AMS Special Session on New Faces in Operator Theory. Seattle, WA
- October 2024 **Analysis and Data Science Seminar,** University at Albany, SUNY. Albany, NY
- October 2024 **AMS Fall Central Sectional Meeting:** Special Session on Multivariable Operator Theory. University at Albany, SUNY. Albany, NY
- September 2024 **Brown University Analysis Seminar.** Providence, Rhode Island.
- March 2024 **Southeastern Analysis Meeting (Plenary).** University of Florida.
- October 2023 **Purdue University Operator Algebras Seminar.**
- August 2023 **Stockholm University Analysis Seminar.**
- August 2023 **IWOTA 2023.** Hilbert function spaces/Model spaces, their operators, and applications (two talks). Helsinki, Finland.
- July 2023 **Saarland University Analysis Seminar.**
- April 2023 **University of Florida Analysis & Probability Seminar (Online).**
- February 2023 **University of South Florida Analysis Seminar.**
- November 2022 **Wabash Mini-Conference (Plenary).** IUPUI.
- October 2022 **Women in Operator Theory Workshop.** University of Memphis.
- October 2022 **Midwestern Workshop on Asymptotic Analysis.** Purdue University Fort Wayne.

- September 2022 **IWOTA 2022.** Model spaces, their operators, and applications. Krakow, Poland.
- March 2022 **Joint Mathematics Meetings.** Recent Progress in Function Theory and Operator Theory (Online).
- December 2021 **Canadian Mathematical Society Winter Meeting.** Approximation Theory in Function Spaces (Online).
- November 2021 **Indiana University Analysis Seminar** (Online).
- October 2021 **Focus Program on Analytic Function Spaces.** The Fields Institute (Online).
- September 2021 **Clemson University Analysis Seminar** (Online).
- September 2021 **IUPUI Operator Theory Seminar.**
- June 2021 **Trends in Operator Theory and its Applications.** University of Lisbon (Online).
- June 2021 **ComPlane: the next generation** (Online).
- April 2021 **University of Toledo Complex Analysis & Operator Theory Seminar** (Online).
- September 2020 **IUPUI Operator Theory Seminar** (Online).
- August 2020 **Operator Theory with its Applications.** University of Florida (Online).
- March 2019 **Butler University Colloquim.**

Other Selected Talks

- January 2026 **University of South Florida Analysis Seminar.** Tampa, Florida.
- June 2023 **Spaces of Analytic Functions and their Operators.** Saarland University, Germany.
- June 2023 **Harmonic and Complex Analysis: Modern and Classical** (in honor of L. Zalcman). Bar-Ilan University, Israel.
- March 2023 **Operator Theory, Free Probability, and Random Matrices** (in honor of H. Bercovici). Georgia Institute of Technology.
- February 2023 **Indiana University Analysis Seminar.**
- July 2021 **Working Groups in Operator Theory.** Lorentz Center (Online).
- March 2021 **The Southeastern Analysis Meeting.** University of Florida (Online).
- February 2021 **Early Career Workshop in Operator Theory/Algebras** (Online).
- December 2019 **Szegő Graduate Seminar.** Washington University in St. Louis.
- June 2019 **Advanced Course in Operator Theory and Complex Analysis.** U. Paris-Est Marne-la-Vallée.
- April 2019 **Washington University in St. Louis Analysis Seminar.**

- December 2018 **Washington University in St. Louis Analysis Seminar.**
- September 2017 **Szegő Graduate Seminar.** Washington University in St. Louis.
- October 2016 **Szegő Graduate Seminar.** Washington University in St. Louis.
- January 2016 **Joint Mathematics Meetings.** AMS Special Session on Functional Analysis and Operator Theory.

Professional Service

Referee for: *Afrika Matematika, Annali dell'Universita di Ferrara, Bulletin of the London Mathematical Society, Complex Analysis and Operator Theory, Constructive Approximation, Demonstratio Mathematica, Linear & Multilinear Algebra (2), Houston Journal of Mathematics, Rocky Mountain Journal of Mathematics.*

Reviewer for: *AMS Mathematical Reviews, Banff International Research Station (Equity, Diversity, & Inclusion)*

Departmental Service

- 2025 – 2027 **Nagle Lecture Series Committee,** *Department of Mathematics & Statistics, University of South Florida.*
- 2022 **Undergraduate Math Club Faculty Advisor,** *Department of Mathematics, Indiana University.*
- 2020 **Director,** *Mathematics and Statistics Online Help Room,* Washington University in St. Louis.
sites.wustl.edu/mathhelp/ 
- 2019 **Committee Member,** *Guido L. Weiss Teaching & Service Award.*
 Selected by department faculty to decide presentation of faculty award for teaching and service
- 2018 **Graduate Council Representative,** *Washington University in St. Louis.*
 Selected by department faculty.

Student Mentorship

- 2023 **Summer Undergraduate Research.** Mentees: Teresa Chen, Thomas Radke, Ben Roark (Coe College). Joint w/ B. Miller. Project: Berezin range of composition operators on H^2 (recently accepted paper to The Rose-Hulman Undergraduate Mathematics Journal).

- 2020 **Undergraduate Teaching Assistant Mentor**, *Calc I/II*, Washington University in St. Louis (8 students).

Community Involvement and Outreach

- 2020 – 2026 **Founding Member/Organizer**, *Operator Theory Talks for Early Researchers (OTTER)*.
<http://www.sites.google.com/view/otter-math> ↗
- 2016 – 2018 **Szegő Graduate Seminar Organizer**, *Washington University in St. Louis*.
Bi-weekly math seminars for graduate students, given by graduate students
- 2017 **Math Circle**, *Washington University in St. Louis*.
Presentation of 'Hilbert's Hotel' to K-12 students.
- 2015 **Presenter, National Chemistry Week and Celebrate Science Indiana**, Indiana State Fairgrounds and Children's Museum of Indianapolis.
Presentation of an electrochemical cell, in the form of a potato battery, to K-12 students.

Conference Organization

- August 2026 **IWOTA Special Session on Invariant Subspaces and Cyclicity**.
2026 IWOTA, Laval University, Quebec.
- January 2026 **AMS Special Session on Recent Endeavors in Complex Analysis and Operator Theory**.
2026 JMM, Washington, D.C.
- January 2025 **AMS Special Session on Recent Advances in Function Theoretic Operator Theory**.
2025 JMM, Seattle.
- January 2022 **The OTTER Math Meeting**.
Online ↗.

Invited Workshop Participation

- October 2022 **Working Groups for Women in Operator Theory**, *University of Memphis* ↗.
- July 2021 **Working Groups for Women in Operator Theory**, *Lorentz Center* ↗ (Online).

June 2017 **Complex Functions, Operators, Partial Differential Equations, and Applications** (in honor of D. Khavinson), *Mittag-Leffler Insitute* .

Professional Memberships

American Mathematical Society

MathSafe (trained) 

Other Professional Experience

Summer 2016 **Science Undergraduate Laboratory Intern**, *U.S. Department of Energy*, Oak Ridge National Laboratory, Mentor: Robert Bridges.
Project: Non-Linear Dimension Reduction

2014 – 2016 **Peer Tutoring Supervisor**, *Engineering Dual Degree Program*, Butler University.

2014 – 2015 **Engineering Intern**, *Freije-RSC Engineered Solutions Co.*, Fishers, Indiana.
Focus: Industrial Refrigeration